GYM ANYTHING FOR ROBOTICS

A physical laundry workcell

Five sewn garments, two standard robots,
camera-based operation and auditable dynamics



Technical and experimental dossier / Newton 1.5.0, Warp 1.16.0

This document describes the implemented digital environment, its numerical mechanisms, parameter basis, component experiments, interaction recordings, graphics, measured runtime, and remote operator interface.

Scope: environment creation, not an autonomous folding solution. Assisted initial placements are explicitly identified. Subsequent robot motion is physical actuator control, not cloth animation. Numerical evidence is distinguished from visual inspection.

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1 Purpose and evidence rules

1.1 What is being built

An operating model must manipulate actual simulated garments with standard robot arms. The scene is not arranged to encode a solution, and success is not triggered by executing a prescribed action sequence. Grasping requires contact; release requires opening the fingers and transferring support to the scene. The environment exposes observations and physical controls so another model can supply task-solving behavior.

Digital-only development is a project constraint, not an unfinished deliverable. Manufacturer documentation, published experiments and defensible engineering estimates supply dimensions and coefficients. We cannot acquire textile specimens or run physical calibration. Accordingly, this report identifies estimates but does not require nonexistent laboratory access for acceptance. Observed implementation defects are not excused as calibration limitations.

Property	Required behavior	Not a completion requirement
Objects	Distinct sewn panels, real openings, material mass and contact	Yarn-level mechanical simulation
Robot	Manufacturer kinematics, inertias, limits, finite contact	Autonomous garment acquisition
Evidence	Component and combined physical interaction, inspected views	Solving every garment task
Sensing	Current, correctly projected RGB and robot encoders	Having the goal visible in every initial camera pose
Realism	Researched values, plausible scene, finite stable dynamics	Physical-world specimen fitting
Performance	Full observed workload below the target ratio	Claiming parallel throughput as single-environment latency

Table 1: Acceptance scope. Assisting a component test is permitted; hiding the assistance is not.

1.2 Scope of the evidence

The P6 towel demonstration uses a VBD cloth model and specialized clamp/roller fixtures. It shows a released half-fold of a 600 by 400 mm, 72 g woven towel. The five-garment P13 track compiles sewn garments for Style3D. The workcell uses those P13 garments and adds two Panda robots, MuJoCo rigid dynamics and explicit robot–cloth coupling.

Those are different inventories and contact paths. A P6 success is not proof that the Panda workcell folds a terry towel, nor is its runtime a camera/history-inclusive performance measurement. Its mechanism experiments and substantial interaction trace set a useful evidence standard. This report explains the workcell's own forces and shows its own interaction.

1.3 Authority and continuity

Policy actions write bounded joint targets. They do not set particle positions, pin cloth to the fingers, change masses or turn collisions off. A privileged teacher can inspect geometry, task gates and per-step history. Diagnostic component initialization may place cloth and robot before tick zero; the initial transform and selected material points are recorded.

Each continuous video has one trajectory with strictly increasing ticks from zero, fixed frame cadence, no checkpoint substitutions and no temporal pose interpolation. Multi-camera panels are simultaneous views of the same tick. Separate trials, failed attempts and causal controls remain separate recordings. Rendering a saved state is not a resumed physical rollout: the renderer advances no simulation time. Its export helper reconstructs the model only to obtain topology and visual metadata; it performs zero physics steps.

1.4 Evidence vocabulary

Source deduction means an equation or control path was read in the pinned implementation. **Measured result** means a named run produced a stored value. **Visual inspection** means the final render or decoded video was actually opened and examined. A successful import, test count or file-existence check does not substitute for any of these.

2 Research basis and design choices

2.1 Different garments require different constructions

Wang et al. report material-dependent, directional stretch and bending response, with different wrinkle patterns even for identical garment cuts [3]. This supports retaining grain frames and separate shear/bending parameters. The present constant-coefficient shell is a reduced implementation; the paper's piecewise-linear fitted constitutive model is not claimed to be implemented.

In this repository, jersey sleeves, poplin cuffs, denim pocket bags and the asymmetric overlay are actual panel groups with authored rest coordinates. Their seams identify selected boundary vertices; openings retain unjoined boundaries. A pocket therefore adds mass and possible self-contact instead of being only a dark patch. The jeans edge experiment demonstrates why this matters numerically: nearby pocket edges can limit a proposal even when the legs are freely hanging. Both the constructed geometry and its dynamics must be examined, not inferred from an inventory name.

2.2 Terry surface, mechanical shell and numerical mesh

Angelova and Kushlova distinguish pile yarn, ground structure, density and pile height in towel design [9]. The digital towel consequently separates raised loops from woven hems and dobby bands. Its authored size is 450 by 700 mm, its areal density is 450 g/m^2, and its unloaded envelope is 3 mm. Those dimensions are the retained construction, not a claim of identity with a particular catalog product. The default render loops remain within that envelope.

Mechanical discretization and visual finish serve different purposes. A 40 mm triangle cannot resolve individual yarn contact. Rendering interpolated normals and material-attached fibres can improve appearance without adding mechanical degrees of freedom. It cannot correct an absent pad witness or make an ungrasped material point follow a robot. The paired normal-map/shading images and the force-linked handling trace test these separate questions.

2.3 Robot and workstation references

The Franka Hand manual specifies 80 mm total stroke and 50 mm/s finger travel [5]. Those limits inform the physical actuator targets; arm geometry, inertias and effort limits come from the manufacturer-description asset. Controller gains and dry pad friction are engineering choices recorded in the mechanics section. Real bimanual work such as SpeedFolding [8] supports using two independently controlled arms over a shared work surface. No published policy success rate or runtime is borrowed as a simulator result.

Evidence source	Decision supported	Implementation check
Manufacturer robot files/manual	Real joints, links, inertia, travel and effort	Asset hashes; realized FK and finger motion
Published cloth mechanics	Material coordinates; directional elastic response	Read native force kernels; inspect active coefficient routing
Terry construction literature	Distinct ground, pile and woven bands	Metric close-up; loops tied to physical triangles
IPC/PBD methods	Crossing guard and local normal-correction candidate	Saved-proposal controls; complete garment release
Real bimanual workstations	Shared table, independent arms, physical camera mounting	Workcell render and collision/geometry inventory

Table 2: Sources justify model and design choices. Measured outcomes come from the named digital experiments, not from the references.

3 Implemented mechanics and units

3.1 Two topologies, one physical garment

Each cut panel has material coordinates u, v in metres, a grain direction, material profile and boundary/seam paths. Sewing identifies selected panel vertices into a welded physical index. Material coordinates remain duplicated across panels so that a pocket and its host retain their own grain frames. The robot and collisions see the resulting physical positions x_i; the elastic calculation also retains the per-triangle material coordinates.

Mass is assembled from material area, areal density and construction layers:

$$m_i = \sum_{t \ni i} \frac{\rho_{A,t} A_t}{3} + m_{i,\text{construction}}, \qquad M = \sum_i m_i. \tag{1}$$

The inventory tables below are generated from the compiler, not inferred from the color of an image. Hems and overlays contribute local construction mass. Sewn boundaries are vertex identification, not breakable stitches. Buttons, zipper teeth and thread lines supply visual construction detail; this backend does not simulate unbuttoning, zipper opening or thread rupture.

3.2 Anisotropic membrane forces

For a triangle define $D_m = [u_1 - u_0, u_2 - u_0]$, where each u_i is a 2-D material coordinate, and $F = [x_1 - x_0, x_2 - x_0] D_m^{-1} = [F_u, F_v]$. The adapter rotates material coordinates so u is weft and v is warp. Let $l_u = \|F_u\|$, $l_v = \|F_v\|$, $\hat{u} = F_u / l_u$ and $\hat{v} = F_v / l_v$. The actual Newton 1.5 stretch kernel accumulates

$$\begin{aligned} f_i = -A_t \big[& k_u (l_u - 1) d_{u,i} \hat{u} + k_v (l_v - 1) d_{v,i} \hat{v} \\ & + k_s (\hat{u} \cdot \hat{v}) (d_{v,i} \hat{u} + d_{u,i} \hat{v}) \big], \end{aligned} \tag{2}$$

where $d_{u,i}, d_{v,i}$ are the corresponding rows/column sums of D_m^{-1}. This is the force expression in `style3d/kernels.py: eval_stretch_kernel`; it is not a claim that every analytical constitutive term in the repository is active. The stretch/shear coefficients have units N/m and are constant per triangle. Profile fields for nonlinear tension stiffening and cross-coupling are not consumed as those additional constitutive terms by this backend.

3.3 Bending

The native bending kernel implements the Crouzeix–Raviart quadratic model [4]. For an interior edge with rest incident area $A_0 + A_1$ and cotangent stencil w,

$$Q = \frac{3}{A_0 + A_1} w^T w, \quad E_b = \frac{1}{2} k_b x^T Q x, \quad f_i = -k_b \sum_j Q_{ij} x_j. \tag{3}$$

The adapter supplies anisotropic panel bending priors, converted into edge coefficients by Newton's builder. Separate panels, seam welding and local mass alter the response. There is no permanent-crease memory, plastic rest-angle update or simulated iron. A fold must persist through gravity/contact after release, not because a renderer stores the desired crease.

3.4 Dissipation and air drag

The inherited membrane-rate kernel supplies Kelvin–Voigt damping using objective stretch rates $\dot{l}_u, \dot{l}_v$ and cosine-shear rate $\dot{c}$, where $c = \hat{u} \cdot \hat{v}$:

$$f_{i,d} = -A_t \left(c_u \dot{l}_u \nabla_i l_u + c_v \dot{l}_v \nabla_i l_v + c_s \dot{c} \nabla_i c \right). \tag{4}$$

All three material rates use the selected profile's membrane damping prior. The bending-rate kernel applies its declared dihedral stencil d_i as $f_{i,d} = -\ell_e c_b \dot{\theta} d_i$ with $\dot{\theta} = \sum_i d_i \cdot v_i$. Its instantaneous work is $\sum_i f_{i,d} \cdot v_i = -\ell_e c_b \dot{\theta}^2 \le 0$ for nonnegative c_b. These kernels add forces; they do not directly damp absolute world velocity or hold the garment in a preferred pose.

For each live triangle, normal relative speed is $s = n \cdot ((v_0 + v_1 + v_2)/3 - v_{wind})$. The total drag is

$$f_{air} = -\tfrac{1}{2} \rho_{air} C_D A |s| s\, n, \quad \rho_{air} = 1.204\,\text{kg/m}^3, \quad C_D = 1.17. \tag{5}$$

One third is applied to each vertex. This is a reduced normal-pressure drag model, not a computational fluid simulation of air or porous textile flow. The workcell has no imposed wind.

3.5 Rigid contact and two-way robot reaction

Native cloth/shape contact uses finite-radius particles, shape geometry, normal penalty/damping and regularized tangential friction. Normal stiffness is 50,000 N/m and damping is 10 N s/m. The coefficient is mixed as $\mu = \sqrt{\mu_{cloth}\mu_{shape}}$. The source function is `evaluate_body_particle_contact` in Newton's VBD rigid kernels; Style3D uses this same function. The bridge does not invent an independent grasp law.

More precisely, for penetration $\delta > 0$, outward normal n and relative translation Δx over a tick, the normal force is

$$f_n^{vec} = k_e \delta n - \frac{k_d}{h} \min(n \cdot \Delta x, 0) n. \tag{6}$$

Let $u = (I - nn^T)\Delta x$, $s = \|u\|$ and $\epsilon = \epsilon_v h$. The friction force is

$$f_t = -\mu k_e \delta \, g(s) u, \qquad g(s) = \begin{cases} (2 - s/\epsilon)/\epsilon, & 0 < s \le \epsilon, \\ 1/s, & s > \epsilon. \end{cases} \tag{7}$$

At zero slip it is zero. Friction uses the elastic normal load $k_e\delta$, not the damping-augmented normal force. This regularization permits small slip; it is not a separate static/kinetic friction projection.

At the final nonlinear contact evaluation, a contact force f_c on the cloth produces the robot wrench

$$W_b = \sum_{c \to b} \begin{bmatrix} -f_c \\ (p_c - r_{COM,b}) \times (-f_c) \end{bmatrix}. \tag{8}$$

The robot consumes this wrench on the *next* tick. Sampling does not sum the same contact force over 20 nonlinear iterations. History exposes both the applied preceding-tick wrench and the newly queued wrench. This is partitioned coupling with a 0.6944 ms delay, not a monolithic coupled solve. Equal-and-opposite sampled forces do not by themselves establish exact discrete-time energy or momentum conservation of that partitioned scheme.

Spatial sampling matters. Rigid contact is sampled at cloth particles; a narrow finger between coarse vertices can visually overlap a triangle while missing a contact witness. A bimanual fixture with the fingers between vertices shows exactly this: zero robot reaction and no cloth lift despite finger closure. A component grip centred on an actual material vertex establishes the implemented force path but does not erase that spatial-resolution limitation. The surface is not secretly attached after a visually plausible grasp.

3.6 Self-contact and finite-table impact

Style3D supplies proximity contact; a repository-owned IPC adapter adds a zero-distance barrier force and continuous crossing protection. Its activation distance is 4 mm, minimum distance zero, barrier prior 10^4, safety factor 0.99 and local exclusion one topological graph hop. Zero minimum distance avoids making welded seams infeasible. Therefore this topology guard is not proof of exact positive-thickness separation at every sewn junction.

IPC checks the proposal before state acceptance. It is an operator split, not an assertion that the native linear solve optimizes the complete IPC incremental potential. Exact nonlocal triangle checks follow each step. Broadphase rejection optimizations are conservative; uncertain pairs retain the original IPC predicates.

An important limitation of a pure operator split is its global acceptance fraction: one closing pocket edge can scale down motion of every cloth point, including remote free legs. The workcell therefore adds an explicit local normal projection before that final safeguard. For a contact stencil at the beginning of a tick, let d be its separation, $g_i = \nabla_i d$, $w_i = 1/m_i$ and Δx_i the proposed displacement. The linearized constraint and correction are

$$C = 0.8 \max(d - 10^{-4}, 0) - 2\,10^{-6} + \sum_i g_i \cdot \Delta x_i \ge 0, \qquad \delta x_i = \frac{-\min(C, 0)}{\sum_j w_j \|g_j\|^2} w_i g_i. \tag{9}$$

This is a mass-weighted normal projection [13], not a garment attachment. It changes only involved stencil vertices. Iteration selects the most violated constraint, with a declared maximum of 128 corrections; it is invoked only when the original admissible CCD fraction is below one. Below 0.1 mm, normal closure is prohibited, with a two-micrometre numerical separation margin (twice the default CCD tolerance). Neither value is asserted to be a measured fabric thickness. Final CCD is run again on the corrected, float32-representable proposal, followed by the unchanged crossing audit. The added impulse $m_i \delta x_i / h$ is separately recorded; residual global clipping is still explicit in the trace. It does not claim to be a monolithic frictional contact solve. This correction is enabled in the workcell, and the garment and

full RGB/history checks reported below use it. An explicit configuration flag runs the path without it for controlled comparison.

The initial collision query is optimized separately from physical acceptance. After a local correction, the next tick first retests the previous edge–edge or face–vertex witness, with its swept bounding boxes and the same IPC narrow phase. A miss falls back to the complete mesh query; it never certifies the whole garment as safe. A hit only establishes that correction is needed. The complete fractional CCD still controls acceptance of the resulting proposal. This reuses search information, not the previous tick's collision verdict.

The local projection also includes the existing finite-table top constraint, using its unchanged particle radius and one-micrometre contact epsilon. It selects the most violated cloth or table constraint on each iteration. A table correction acts along the vertical normal of the static support; it is an external reaction, not a momentum-conserving cloth–cloth impulse. The cloth-only candidate in the contact controls below omits this coupling and moves a particle 37 micrometres into the table after the table guard has run. The projection therefore revalidates both domains before acceptance.

IPC distance routines return squared distance [14]; displayed distance in metres applies the square root. Correct diagnostic units do not alter the computed barrier or CCD.

The support table is a finite solid, not an infinite plane. Ordinary contact handles sides and underside. A swept-particle fallback handles crossing of its finite top face and removes inward normal velocity with zero restitution. It leaves tangential state to the contact solver. The top-face verification domain compares distances to all box faces, so a valid overhang or cloth below the underside is not rejected simply for being below the top height.

3.7 Saved-proposal causal diagnosis

The jeans diagnosis retains the actual native proposal arrays, not merely a last-frame image. Its limiting right-pocket edge pair is $(753, 770)$ against $(752, 787)$. The measured starting gap is $1.4743\,\mu$m; the first-order normal gap change over the proposed tick is -0.753687 mm. The complete baseline CCD query accepts only 0.0009765625 of the proposal, and that same fraction scales the entire garment's displacement. This explains why remote legs can appear frozen even though the colliding stencil is near the pocket.

Identical saved proposal, changed CCD query only	Accepted fraction
Tight inclusion, tolerance 10^{-6}	0.0009765625
Tight inclusion, tolerance 10^{-7}	0.0012817383
Tight inclusion, tolerance 10^{-8}	0.0013542175
Tight inclusion, tolerance 10^{-9}	0.0013604164
Additive CCD	0.0014563348

Table 3: The small accepted fraction persists on the identical arrays when CCD tolerance/method changes, so tolerance alone does not fix it. This is not a new physical garment run.

The complete edge experiments below establish the behavior after local correction; a successful saved-proposal calculation alone is not a runtime qualification.

3.8 Robot dynamics and controller

The two Panda descriptions are hash-verified manufacturer-distributed URDF assets. Link collision/visual meshes, inertia, travel and effort limits are retained. The bases are fixed to physical mounting structures at $(0, \pm 0.78, 0.78)$ m. MuJoCo advances finite-mass articulated dynamics. Robot gravity compensation passes through the actuator sum and effort limits; gravity on the garment is unchanged.

The arm drive uses position stiffness 600 and damping 45; fingers use 1600 and 12 in the corresponding joint SI units. Arm effort limits are those of the asset (87/12 Nm joint groups); the retained finger limit is 20 N per joint. The gripper opens 0–40 mm per finger, with target travel limited to 50 mm/s per finger from the Franka Hand manual [5]. These are physical position servos, not force-controlled acquisition of an arbitrary garment.

Full-workcell numerical setting	Effective value
Engine / Warp	Newton 1.5.0 / Warp 1.16.0
Physics timestep	1/1440 s; 1440 actual physics ticks/s
Cloth nonlinear / linear iterations	20 / 60
Rigid solver iterations	30, implicit-fast integration
Physical garment target edge	40 mm; retained P13 topology
Cloth–shape stiffness / damping	50,000 N/m / 10 N s/m
Contact buffer capacity	8192
Robot reaction timing	Last nonlinear evaluation, applied next tick
Policy images / teacher samples	Requested camera cadence / every physics tick

Table 4: Numerical settings are distinct from textile material values. GPU graph capture does not reduce these counts.

4 Garment construction and parameter ledger

The following inventory is generated from the current garment compiler and retained P13 reset arrays. Axes are metric. Colored panels denote separate cut pieces; this is a diagnostic topology drawing, not a camera observation.

4.1 Cotton-jersey half-sleeve T-shirt

5 panels; 13 declared seams; 4 openings; 650 physical particles and 1,248 triangles. Assembled mass 126.39 g, including local construction mass.

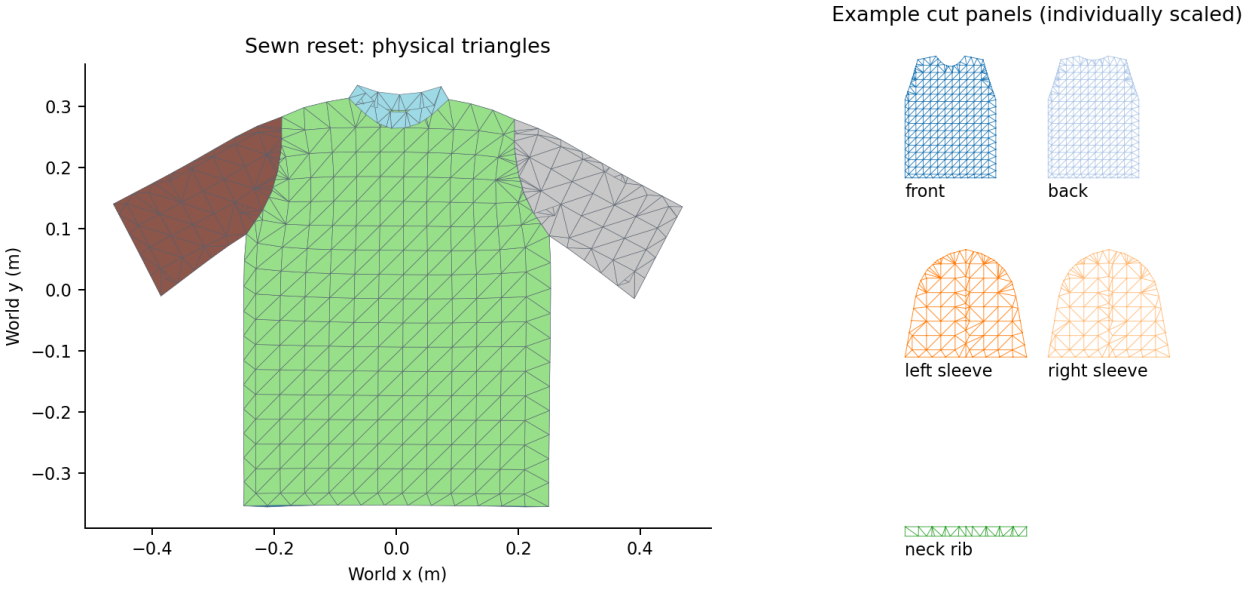


Figure 1: Cotton-jersey half-sleeve T-shirt: retained material topology and reset projection. Atlas panels are separate diagrams, not an exploded simulation state.

Panel	Material	g/m^2	mm
front	cotton jersey	160	0.5
back	cotton jersey	160	0.5
left sleeve	cotton jersey	160	0.5
right sleeve	cotton jersey	160	0.5
neck rib	cotton rib	220	0.8

Openings: neck, left arm, right arm, waist. **Local constructions:** folded neck rib, body hems, sleeve hems.

4.2 Woven cotton button-down shirt

11 panels; 25 declared seams; 6 openings; 1,104 physical particles and 2,114 triangles. Assembled mass 151.16 g, including local construction mass.

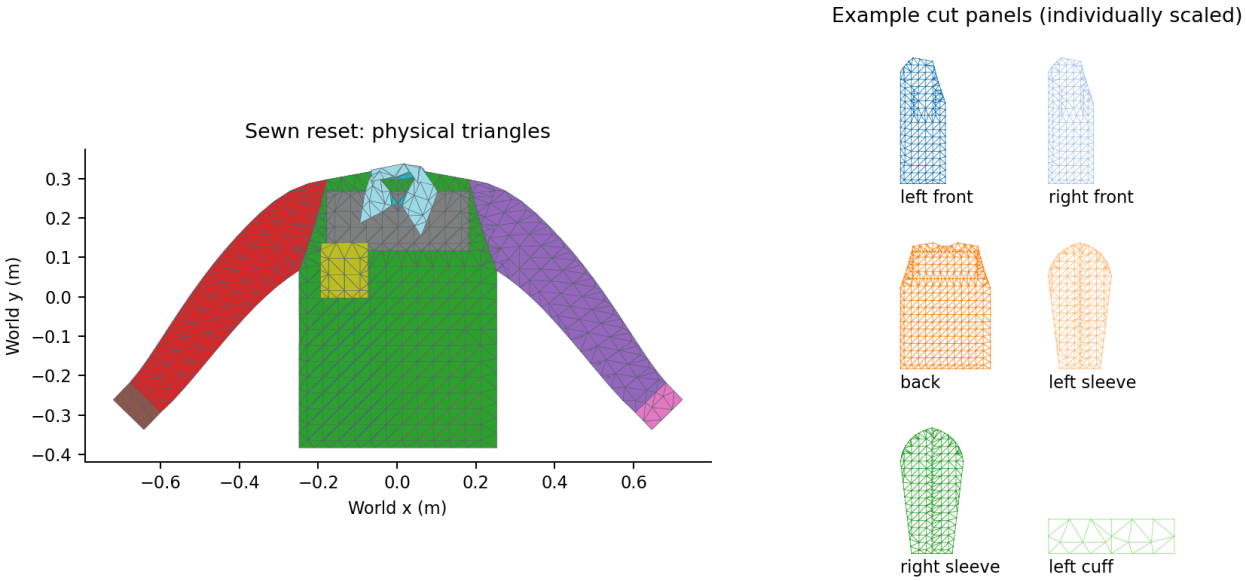


Figure 2: Woven cotton button-down shirt: retained material topology and reset projection. Atlas panels are separate diagrams, not an exploded simulation state.

Panel	Material	g/m^2	mm
left front	cotton poplin	120	0.3
right front	cotton poplin	120	0.3
back	cotton poplin	120	0.3
left sleeve	cotton poplin	120	0.3
right sleeve	cotton poplin	120	0.3
left cuff	cotton poplin	170	0.55
right cuff	cotton poplin	170	0.55
back yoke	cotton poplin	240	0.6
chest pocket	cotton poplin	135	0.42
collar stand	cotton poplin	210	0.7
collar	cotton poplin	220	0.75

Openings: front placket, neck, left wrist, right wrist, chest pocket, waist. **Local constructions:** left placket, right placket, button row, buttonhole row, chest patch pocket, doubled yoke, interfaced collar, interfaced cuffs.

4.3 Five-pocket denim jeans

19 panels; 44 declared seams; 9 openings; 991 physical particles and 1,912 triangles. Assembled mass 357.30 g, including local construction mass.

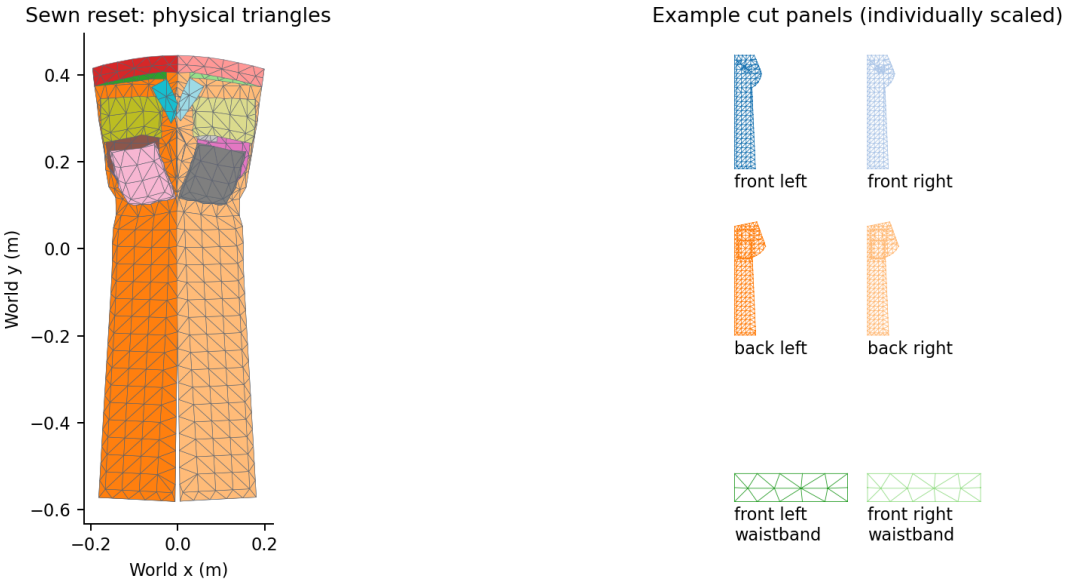


Figure 3: Five-pocket denim jeans: retained material topology and reset projection. Atlas panels are separate diagrams, not an exploded simulation state.

Panel	Material	g/m^2	mm
front left	denim twill	400	0.9
front right	denim twill	400	0.9
back left	denim twill	400	0.9
back right	denim twill	400	0.9
front left waistband	denim twill	650	1.6
front right waistband	denim twill	650	1.6
back left waistband	denim twill	650	1.6
back right waistband	denim twill	650	1.6
left pocket outer	cotton poplin	95	0.25
left pocket inner	cotton poplin	95	0.25
right pocket outer	cotton poplin	95	0.25
right pocket inner	cotton poplin	95	0.25
left back patch pocket	denim twill	430	1.1
right back patch pocket	denim twill	430	1.1
coin pocket	denim twill	430	1.1
back left yoke	denim twill	440	1.05
back right yoke	denim twill	440	1.05
fly shield	denim twill	720	1.8
fly facing	denim twill	650	1.6

Openings: fly, waist, left leg, right leg, left front pocket, right front pocket, left back pocket, right back pocket, coin pocket. **Local constructions:** four piece waistband, front pocket bags, back patch pockets, coin pocket feature, back yokes, zip fly, waist button, belt loops, leg hems.

4.4 Loop-pile terry cotton bath towel

1 panels; 0 declared seams; 1 openings; 273 physical particles and 484 triangles. Assembled mass 141.75 g, including local construction mass.

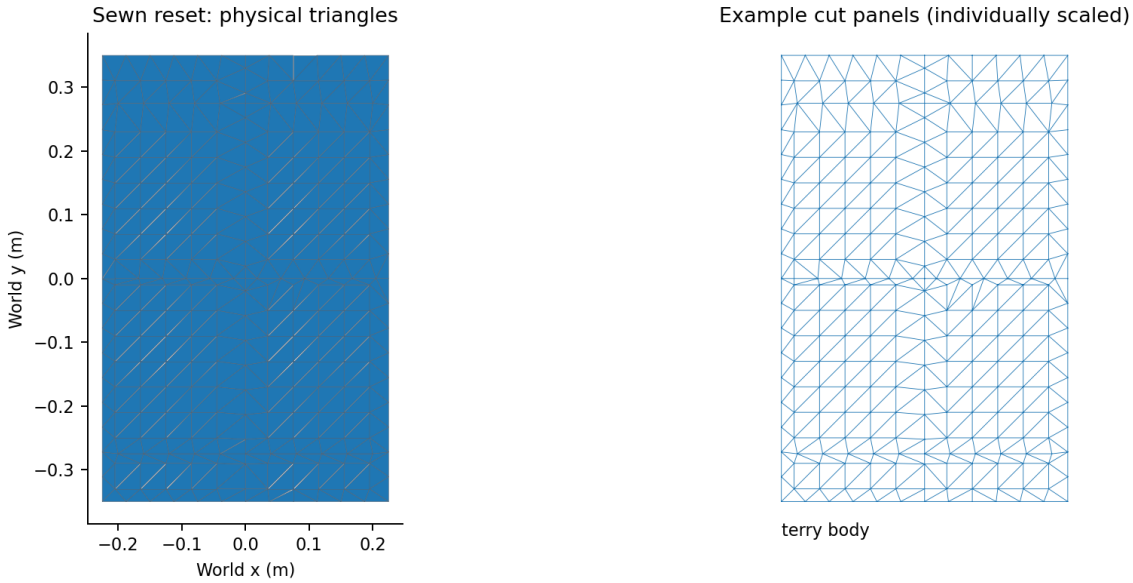


Figure 4: Loop-pile terry cotton bath towel: retained material topology and reset projection. Atlas panels are separate diagrams, not an exploded simulation state.

Panel	Material	g/m^2	mm
terry body	terry cotton	450	3

Openings: perimeter. **Local constructions:** four multilayer hems, bottom dobby band, top dobby band.

4.5 Asymmetric draped one-bell-sleeve top

9 panels; 18 declared seams; 5 openings; 976 physical particles and 1,788 triangles. Assembled mass 150.37 g, including local construction mass.

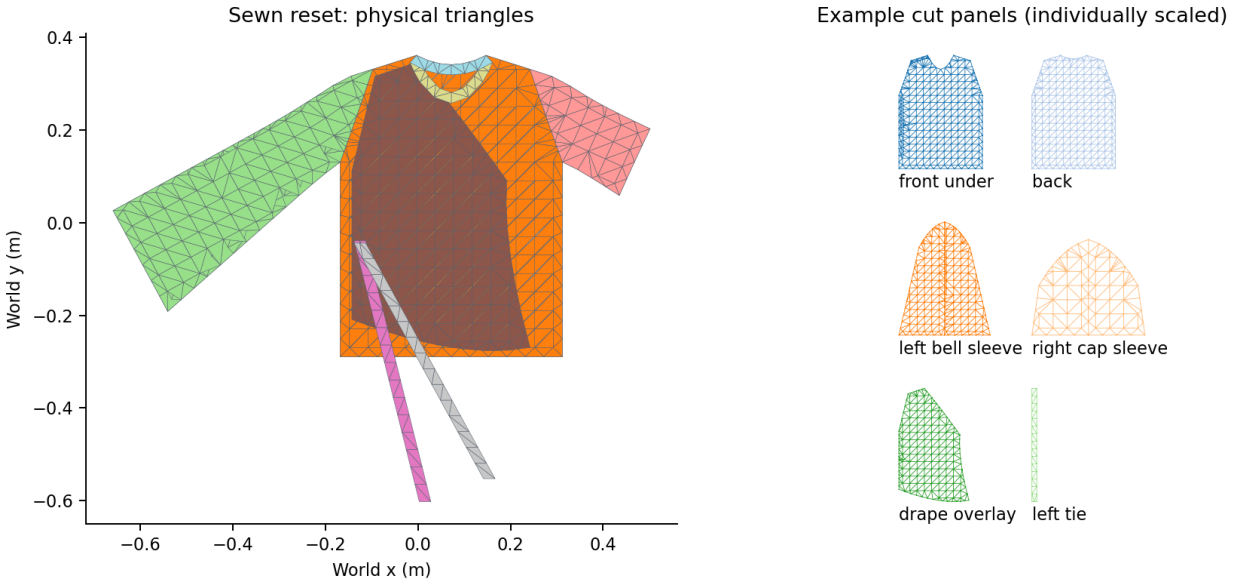


Figure 5: Asymmetric draped one-bell-sleeve top: retained material topology and reset projection. Atlas panels are separate diagrams, not an exploded simulation state.

Panel	Material	g/m^2	mm
front under	viscose crepe	140	0.35
back	viscose crepe	140	0.35
left bell sleeve	viscose crepe	140	0.35
right cap sleeve	viscose crepe	140	0.35
drape overlay	viscose crepe	140	0.35
left tie	viscose crepe	180	0.55
right tie	viscose crepe	180	0.55
front neck facing	viscose crepe	210	0.7
back neck facing	viscose crepe	210	0.7

Openings: neck, left arm, right arm, waist, drape front. **Local constructions:** bias drape, waist ties, neck facing feature, asymmetric sleeves.

4.6 Active material-family coefficients

These are inherited engineering priors informed by the material-family literature, not fitted specimens. The table lists coefficients consumed by the current backend; profile-only nonlinear and compression fields are not represented as active mechanics.

Material	k_u N/m	k_v N/m	k_s N/m	$k_{b,warp}$ N m
cotton jersey	700	500	80	2e-06
cotton poplin	4000	6000	300	8e-06
cotton rib	1200	800	90	3e-06
denim twill	10000	14000	1000	0.00012
terry cotton	3500	4500	300	3.5e-05
viscose crepe	1500	1800	120	3e-06

Table 10: Grain-aligned elastic priors supplied to the native triangle/edge builder.

Material	$k_{b,weft}$ N m	$k_{b,bias}$ N m	c_m N s/m	c_b N m s
cotton jersey	2.5e-06	1.5e-06	0.035	2e-07
cotton poplin	5e-06	4e-06	0.04	5e-07
cotton rib	4e-06	2e-06	0.05	4e-07
denim twill	9e-05	6e-05	0.08	4e-06
terry cotton	3e-05	2.2e-05	0.075	2.5e-06
viscose crepe	2.5e-06	1.8e-06	0.045	3e-07

Table 11: Remaining directional bend and material-rate damping values. All units refer to the adapter inputs, not a volumetric Young's modulus.

Contact coefficient routing. The workcell uses the first panel's table-kinetic prior as its single cloth/shape friction input: 0.36 (tee), 0.33 (shirt), 0.44 (jeans), 0.49 (towel), and 0.37 (asymmetric). Native shape mixing then applies. Separate static, cloth–cloth and finger fields in the richer profile are not independent laws consumed by this adapter.
Parameter scope. Rest-panel dimensions, areal density and grain remain construction-specific. Constant elastic coefficients and damping are inherited material-family engineering estimates. No new values were fitted for Newton 1.5. The profile's nonlinear tension-stiffening, cross-coupling and pressure-compression fields are retained metadata, not additional active constitutive terms.

5 Experimental design and interpretation

5.1 Why several kinds of evidence are needed

An idle finite-state check cannot establish a grasp, and a grasp cannot establish contact with a table edge. The experiments below separate native robot tracking, cloth contact, gravity release, layer separation, construction, camera freshness, history integrity and sustained runtime. Joint unit tests remain useful for rejecting invalid requests; they are not the basis for claiming textile realism.

Experiment	Evidence needed	What it does not establish
Two-arm handling	Cloth follows both hands, nonzero reaction, support transfer, release	Autonomous pickup or general folding
Single-hand grip	Opposing pad contacts, lift and loss of support on opening	Arbitrary contact-point coverage
Table-edge drape	Supported region and downward overhang, finite table geometry	Full cloth/edge continuous collision for every motion
Two-layer release	Gravity-driven settling and no confirmed nonlocal crossing	Yarn compression or crease memory
Garment release	Distinct garment motion, seams/openings retained, finite state	Every possible interaction with that garment
Graph comparison	Same settings and arrays, numerical agreement	Correct material priors by itself
HTTP trial	Real requests, fresh RGB, role restrictions, recorded action	Internet security without a deployment boundary

Table 12: Mechanism-specific evidence prevents a collection of unrelated passes from being called a complete interaction demonstration.

5.2 Bimanual component protocol

The initial towel is rigidly rotated into a hanging configuration between the two hands. Its rest metric, material coefficients, particle masses and dimensions are unchanged. The initial finger openings are 2 mm per side. The protocol closes both hands, raises their target TCPs by 60 mm, holds, lowers onto the table, opens one hand, lowers the remaining edge, opens the other hand, retracts and settles. Actual TCPs and material-point trajectories are measured independently of targets.

A request to move a hand is not evidence that the cloth followed it. The plots therefore compare tracked material-point heights with realized FK TCP height, show actual finger openings, and include contact reactions and table-supported particle counts. The recording retains full-resolution history; displayed curves identify their sampling cadence. Failed trials are not removed from the experiment index or spliced into a successful video.

The successful combined recording does not demonstrate uninterrupted retention through every commanded phase. During lowering, the right tracked point slips out before the command to open that hand. The remaining hand and table carry the cloth, and final release occurs without attachment. This observed slip is part of the recording, not removed or credited as a precisely timed release.

5.3 Single-hand, edge and layer trials

The 2.6667 s one-hand trial measured 48.02 mm cloth lift, 156.63 mm drop after release, sampled peak reaction 2.659 N and zero final nonlocal crossings. Contact witnesses identify opposing pads; the final supported state has no hand reaction. This establishes a narrow physical grasp and release, not the larger bimanual evidence requirement.

The edge trial measured 192.01 mm downward motion of its lowest point with 169 vertices still supported. The layer trial ended with a 6.159 mm vertical extent and 273 supported vertices. Each has 2880 ticks, two simulated seconds and complete per-step history. Their distinct initial conditions are explicitly assisted; they are not consecutive stages of a fold.

5.4 P1–P13 cloth studies

The P1–P13 studies contain gravity/ballistic comparisons, layer-contact controls, free bending release, friction breakaway, zero-friction grasp controls, clamp/roller half-fold, garment construction and exact intersection studies. They are not Newton 1.5 Panda results. In particular, the Style3D contact path does not inherit a P1–P13 fixture's set-valued static friction claim merely because both use the word "friction."

Re-executed, the P6 baseline is finite and released but has 20 confirmed crossings, so its accepted result is not quoted on its own.

5.5 Performance method

Report $R = T_{wall}/T_{sim}$, with a target below 100 for this environment. Startup and first observation are separate from steady rollout cost. The record must state batch size, actual camera count/resolution/samples/cadence, physics iteration counts, and whether teacher history is enabled. An offline high-sample video export is not the latency of the live policy observation. Parallel throughput, if measured, must be reported separately.

6 Recorded workcell results

6.1 Two-arm handling and support transfer

The uninterrupted run contains 11,520 physics ticks (8 simulated seconds). Actual material-point lifts were 58.84 mm (right) and 58.84 mm (left). Final RMS cloth speed was 0.00025 m/s. These are physical component measurements, not a fold-success verdict.

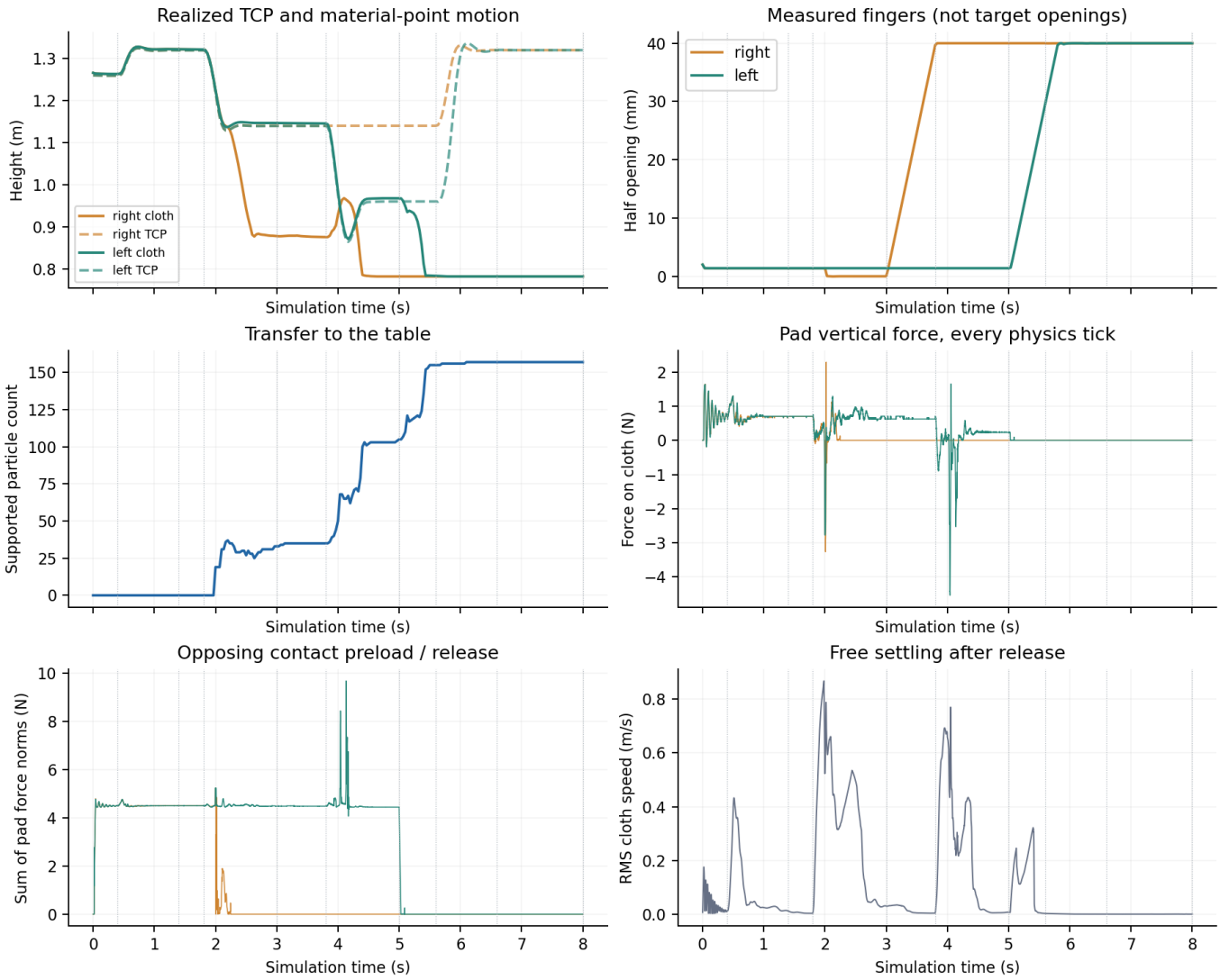


Figure 6: Combined interaction. Top plots use the 30 Hz trajectory cadence; contact forces and cloth speed use every physics tick. No moving-average smoothing is applied.

Phase	Start (s)	End (s)
close both	0.00	0.40
lift both	0.40	1.40
hold two arm support	1.40	1.80
lower to table	1.80	3.00
transfer to left hand and table	3.00	3.80
lower remaining edge	3.80	5.00
release left	5.00	5.60
retract clear	5.60	6.60
settle on table	6.60	8.00

Table 13: Recorded action boundaries, not inferred from video frames.

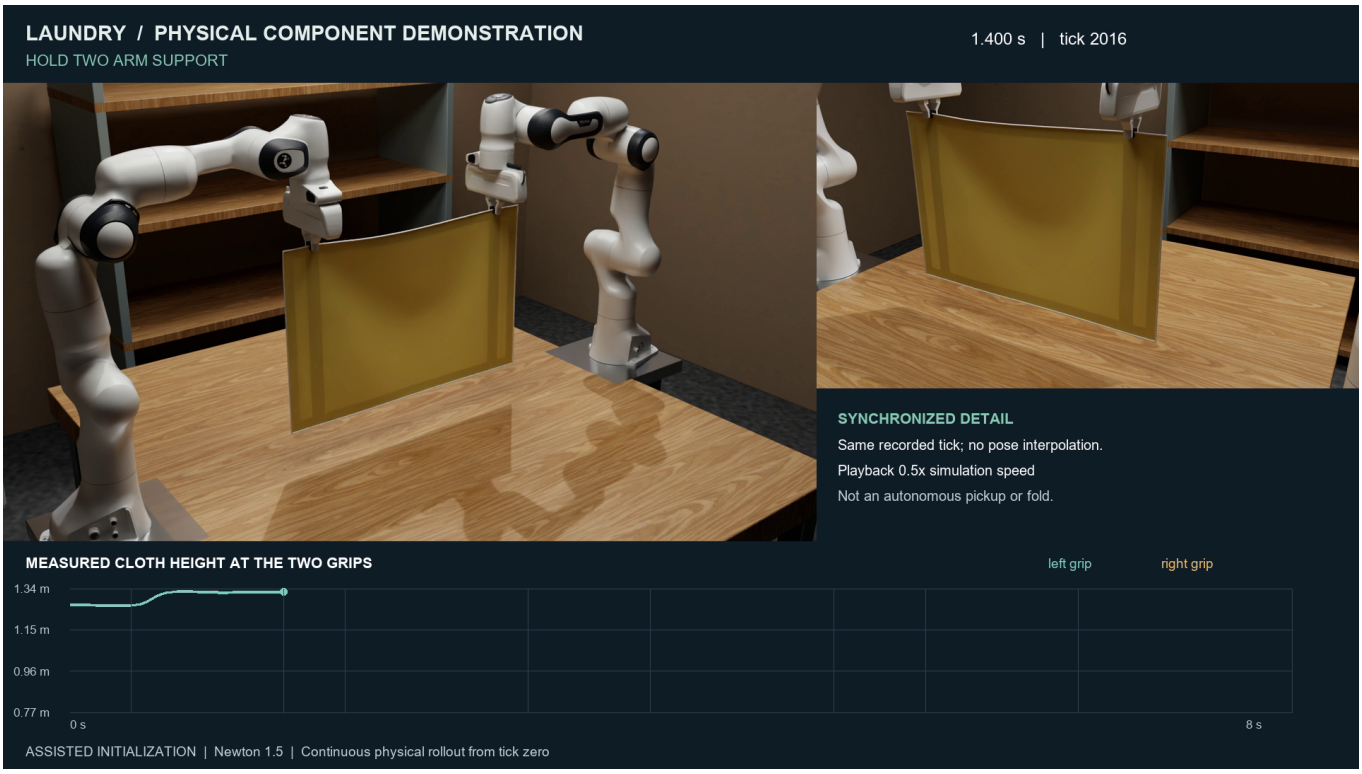


Figure 7: Decoded MP4, tick 2016: both tracked material points lifted with physical grippers. Assisted initial placement; no running attachments.

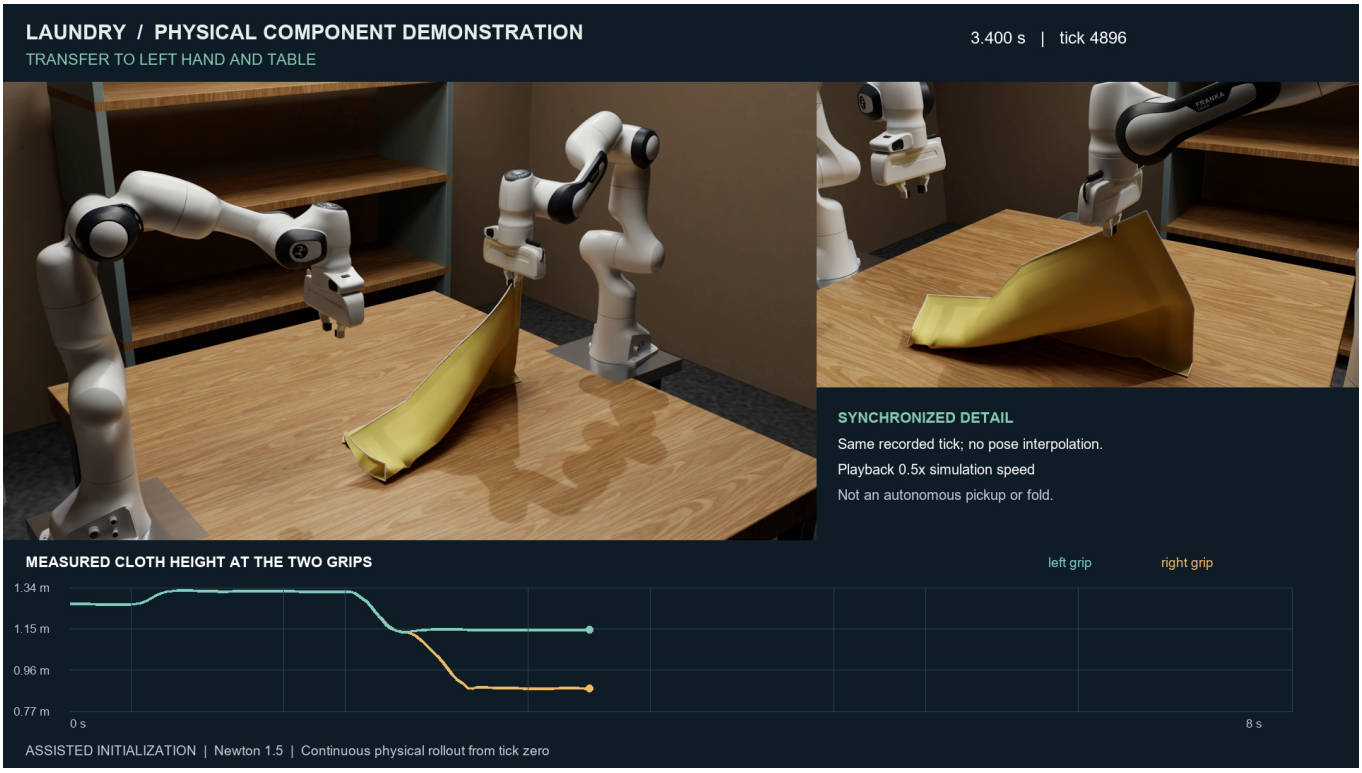


Figure 8: Decoded MP4, tick 4896: support transfers to the table. The right point slipped before the opening command; the phase label names the command, not a retention verdict.

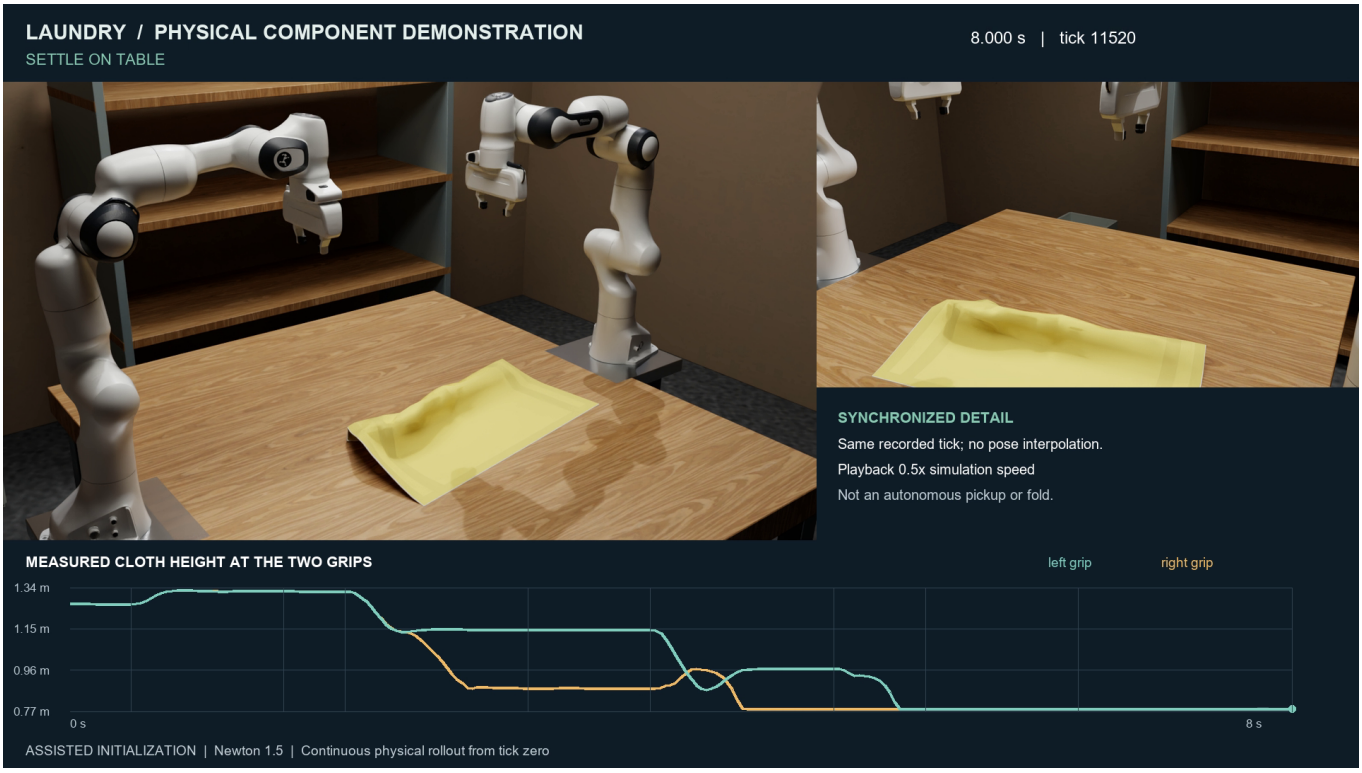


Figure 9: Decoded MP4, tick 11520: both hands are open and the cloth rests on the table. The complete uninterrupted trajectory ends here. This is not a folding-success result.

Component gate	Result
all phases	pass
no error	pass
no crossings	pass
no runtime pose writes	pass
both material points lifted	pass
both fingers open final	pass
table support final	pass
two pad witnesses each grip	pass
all physics ticks recorded	pass
no pad force final	pass
one hand support transfer	pass
all action traces complete	pass

Table 14: Gate outcomes computed from the actual recorded state and contact history.

6.2 towel: table-edge component

2,880 physics ticks; 169 supported particles; lowest-point displacement 192.01 mm; 0 final confirmed nonlocal crossings. This is a separate gravity/contact trial of the complete sewn garment. Principal-stretch extrema are [1.000, 1.000] initially and [0.995, 1.006] finally; final mean absolute area strain is 0.03 percent. A drape pass checks gravity, support and crossing integrity; it is not a pass of the settled-fold strain gates.

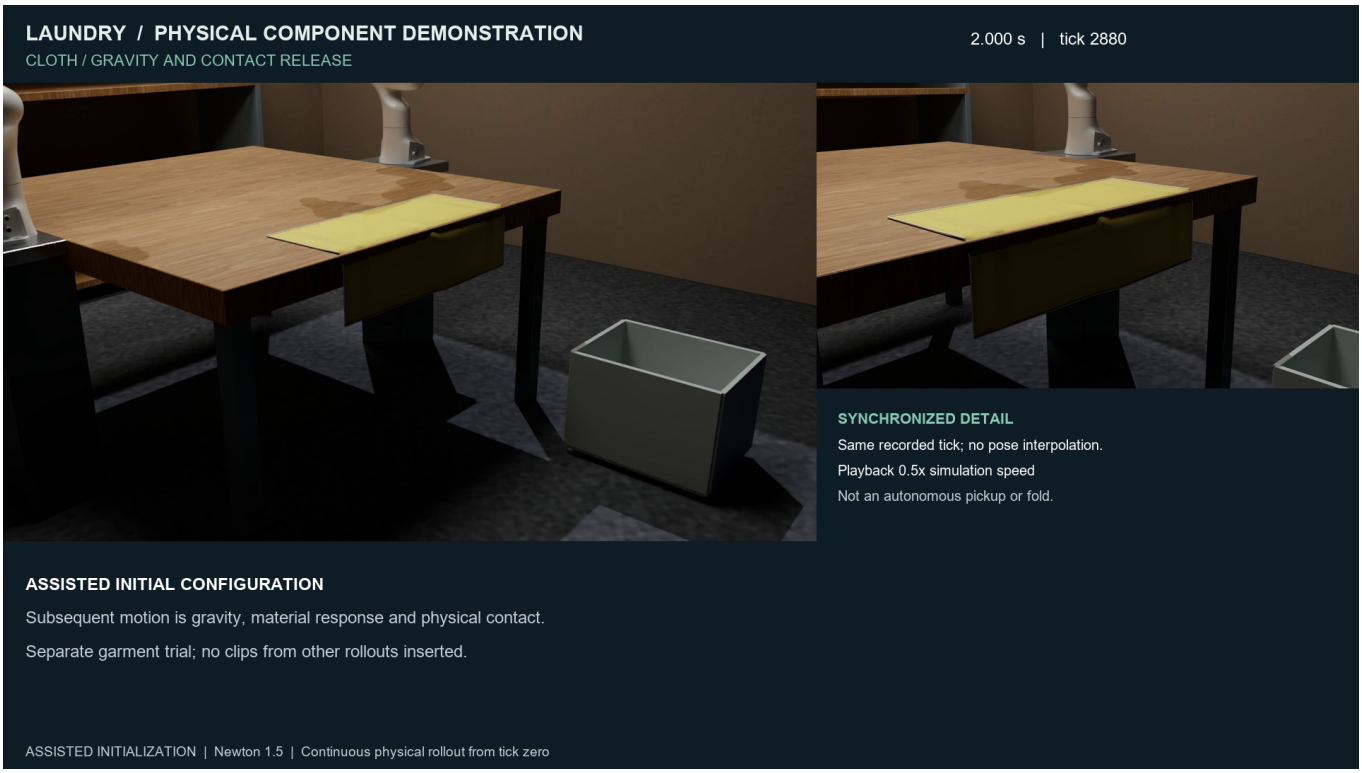


Figure 10: Towel, tick 2880: Newton 1.5 edge trial rendered with the terry pile and synchronized views. The trial had no global CCD clips, so the conditional local correction was not needed. Separate from the bimanual trajectory.

6.3 tee: table-edge component

2,880 physics ticks; 137 supported particles; lowest-point displacement 415.97 mm; 0 final confirmed nonlocal crossings. This is a separate gravity/contact trial of the complete sewn garment. Principal-stretch extrema are [0.695, 1.170] initially and [0.882, 1.101] finally; final mean absolute area strain is 0.66 percent. A drape pass checks gravity, support and crossing integrity; it is not a pass of the settled-fold strain gates.

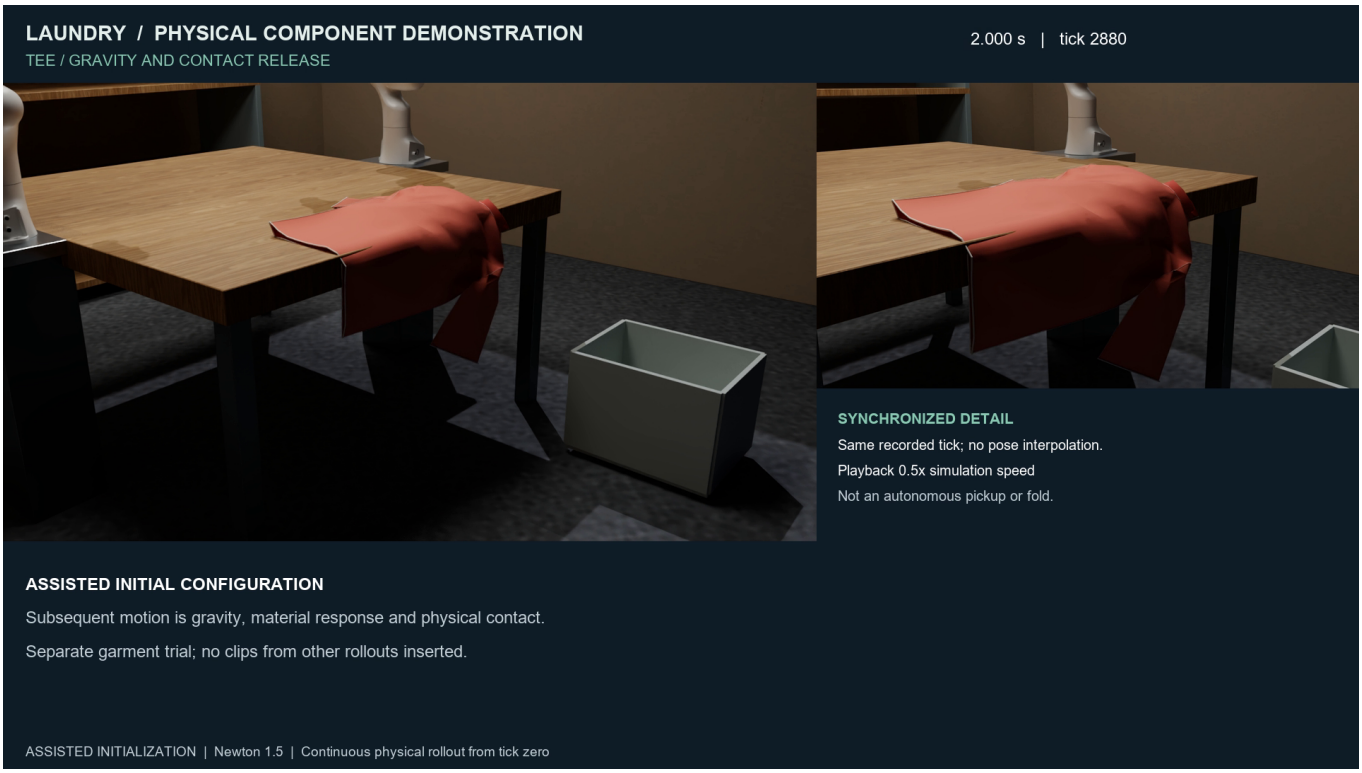


Figure 11: T-shirt, tick 2880: supported body and gravity-driven overhang, with the local contact correction enabled. Separate uninterrupted two-second component trial.

6.4 shirt: table-edge component

2,880 physics ticks; 278 supported particles; lowest-point displacement 715.53 mm; 0 final confirmed nonlocal crossings. This is a separate gravity/contact trial of the complete sewn garment. Principal-stretch extrema are [0.697, 1.232] initially and [0.907, 1.094] finally; final mean absolute area strain is 0.27 percent. A drape pass checks gravity, support and crossing integrity; it is not a pass of the settled-fold strain gates.

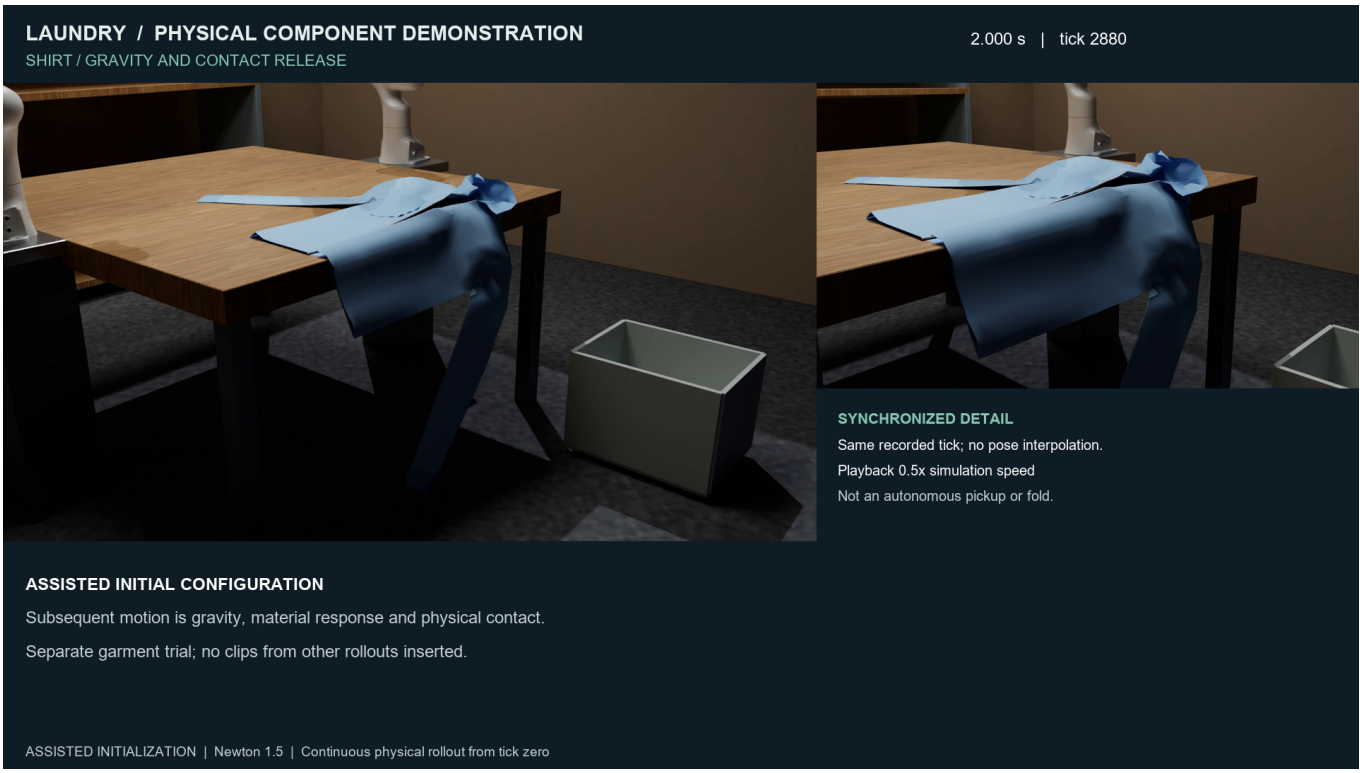


Figure 12: Shirt, tick 2880: long-sleeved sewn construction dropping over the finite support edge. Both views use exactly the same recorded tick.

6.5 jeans: table-edge component

2,880 physics ticks; 89 supported particles; lowest-point displacement 78.28 mm; 0 final confirmed nonlocal crossings. This is a separate gravity/contact trial of the complete sewn garment. Principal-stretch extrema are [0.738, 1.251] initially and [0.503, 1.169] finally; final mean absolute area strain is 0.57 percent. A drape pass checks gravity, support and crossing integrity; it is not a pass of the settled-fold strain gates.

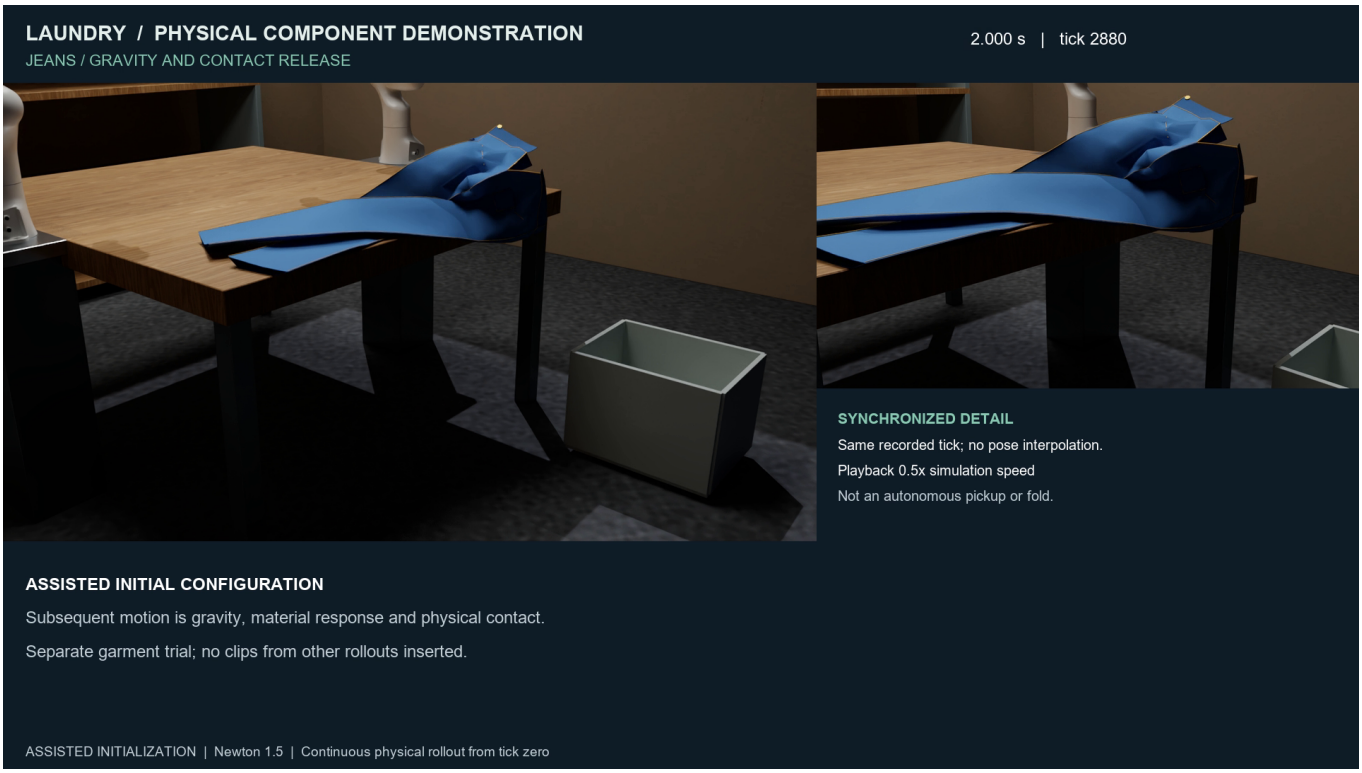


Figure 13: Jeans, tick 2880: supported legs and displaced waist/pocket region. The coarse sewn mesh retains local compressed triangles, disclosed in the strain ledger; this is not a settled-fold pass.

6.6 asymmetric: table-edge component

2,880 physics ticks; 278 supported particles; lowest-point displacement 433.11 mm; 0 final confirmed nonlocal crossings. This is a separate gravity/contact trial of the complete sewn garment. Principal-stretch extrema are [0.720, 1.478] initially and [0.900, 1.358] finally; final mean absolute area strain is 0.25 percent. A drape pass checks gravity, support and crossing integrity; it is not a pass of the settled-fold strain gates.

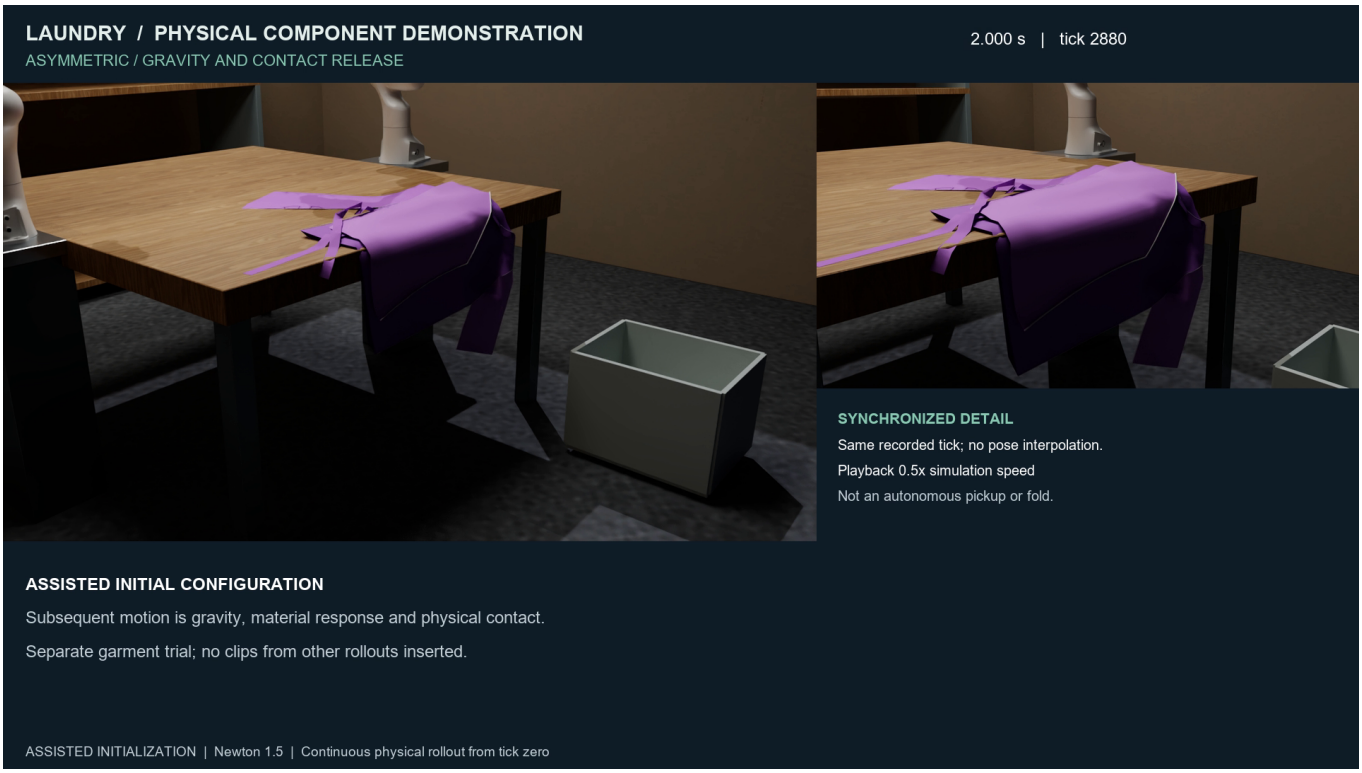


Figure 14: Asymmetric top, tick 2880: overlay and ties deform under gravity while part of the garment remains supported. No robot pickup or task solution is claimed.

6.7 Whole-garment acceptance versus local contact

Every curve below uses a separate recorded table-edge trial. The initial rigid translation, garment, mesh, timestep and iteration counts are retained within each comparison. The vertical axis is the displacement of the lowest physical particle, not a commanded target. Failed candidates remain in the table; a short terminated run is not extended to the planned duration.

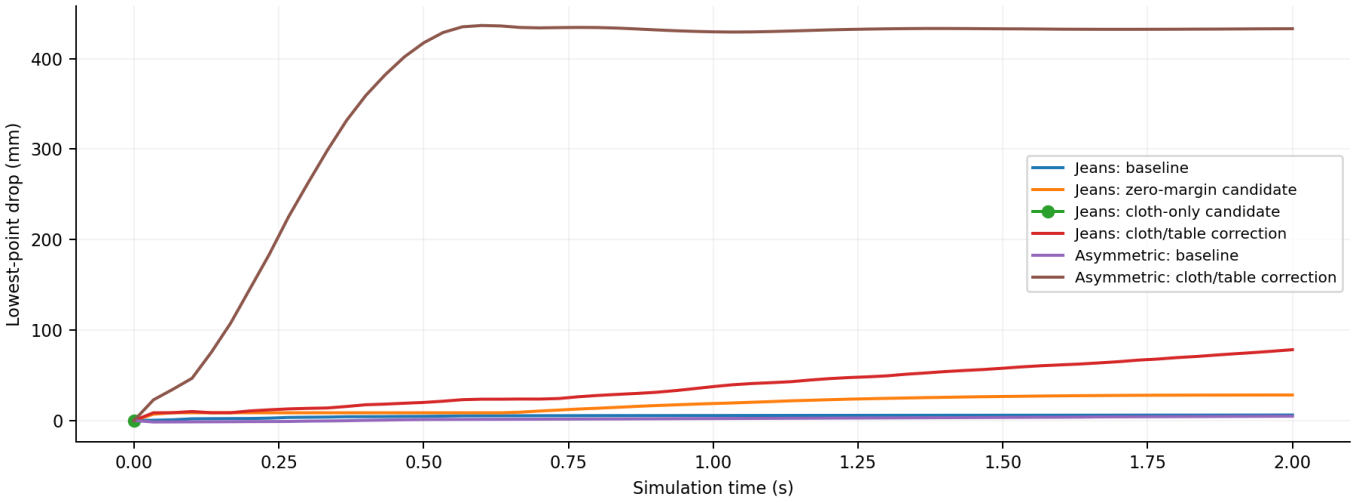


Figure 15: Raw recorded 30 Hz geometry. Separate trials are compared on common axes, not concatenated into a continuous trajectory.

Trial	End tick	Result	Recorded diagnostic
Jeans: baseline	2880	FAIL	supported 1; crossings 0
Jeans: zero-margin candidate	2880	FAIL	supported 54; crossings 0
Jeans: cloth-only candidate	20	FAIL	RuntimeError: Table-plane CCD cannot start from a penetrating accepted state (minimum surface clearance -3.73434067e-05 m)
Jeans: cloth/table correction	2880	pass	supported 89; crossings 0
Asymmetric: baseline	2880	FAIL	supported 5; crossings 0
Asymmetric: cloth/table correction	2880	pass	supported 278; crossings 0

Table 15: Unsuccessful runs are part of the mechanism evidence. End tick is the recorded stop, not a fabricated completion.

6.8 Full-observation performance

One environment per measurement; 1440 actual physics ticks, 30 returned 1280 by 720 policy RGB frames at eight path samples, and every physics tick recorded. Initialization and first observation are excluded from rollout latency and listed separately.

Garment	Wall/sim	RGB (s)	Build (s)	First views (s)
towel	59.88	19.65	8.08	25.29
tee	63.77	15.49	9.23	24.80
shirt	92.79	26.18	11.46	25.95
jeans	99.05	25.89	10.79	26.08
asymmetric	83.05	15.37	11.62	25.16

Table 16: Measured wall-clock seconds per simulated second, not aggregate parallel throughput.

6.9 Actual HTTP integration checks

These requests used the real shared gateway, physical robot backend and separately authenticated roles.

Executed check	Result
role help	pass
role help	pass
privileged state	pass
student state denied	pass
student geometry denied	pass
privileged sewn geometry	pass
student observation	pass
no cloth ground truth in student observation	pass
student RGB fetch	pass
dead renderer recovery retains the identical live physical state	pass
recovered policy RGB fetch	pass
action and credential-scoped idempotency	pass
student result has no privileged reward	pass
student history denied	pass
privileged history	pass
complete per-tick cloth and robot history	pass
student trace denied	pass
invalid target rejected before physics	pass
owner reconciliation without movement	pass
health	pass
left IK/FK matches native robot frame	pass
right IK/FK matches native robot frame	pass
student closes physical fingers	pass
student opens physical fingers	pass
explicit reset	pass
reset changes generation	pass
stale state/image pair rejected	pass

7 Scene, textile appearance and video evidence

7.1 A physically represented workstation

The 1.6 by 1.4 m folding table stands 0.78 m above the floor. Robot pedestals, mounts, table, storage furniture, baskets, walls and floor have corresponding collision geometry. The standard two-arm arrangement is informed by real bimanual garment workstations such as SpeedFolding [8], not by adding task-specific invisible supports. Material finish uses wood, wall, floor and textile texture maps, area illumination and an environment light.

7.2 Surface geometry is tied to the simulation

Each render panel samples physical triangles through fixed barycentric weights b: $x_{render} = \sum_{i=0}^{2} b_i x_i$. Material UVs and construction lines use the same panel mapping. No animation curve drives a fold. Render-only panels retain a finite-thickness shell; visual buttons and stitches follow their material attachments rather than remaining in the reset position.

Calculating normals after subdividing coarse triangles would let interior child vertices see only their parent face, which produces patches of shading across a supposedly smooth textile. The renderer therefore interpolates normals from the original panel vertices before normalization. Its paired diagnostic keeps physical coordinates and the finite-thickness shell points unchanged. Smoother shading is not new physical resolution; coarse silhouettes and between-vertex contact remain limitations.

7.3 Terry pile and construction finish

A plain terry normal map, with its regular oval pattern and shallow surface, reads poorly as loop pile. A render-only yarn-loop treatment is therefore enabled for towel rendering within the declared 3 mm textile envelope. The base is recessed and deterministic, non-identical loops follow actual material triangles on both sides. It neither adds collision forces nor changes garment thickness in the mechanical model. Woven hems and dobby bands remain distinct from pile.

Published terry studies identify pile yarn, ground yarn, weave density and loop height as distinct construction variables [9]. They justify representing a textured raised pile and woven borders, not claiming that our particular digital loop spacing or height was measured on a named product. The digital visual scale is constrained by this asset's existing dimensions and thickness; it is a documented appearance estimate.

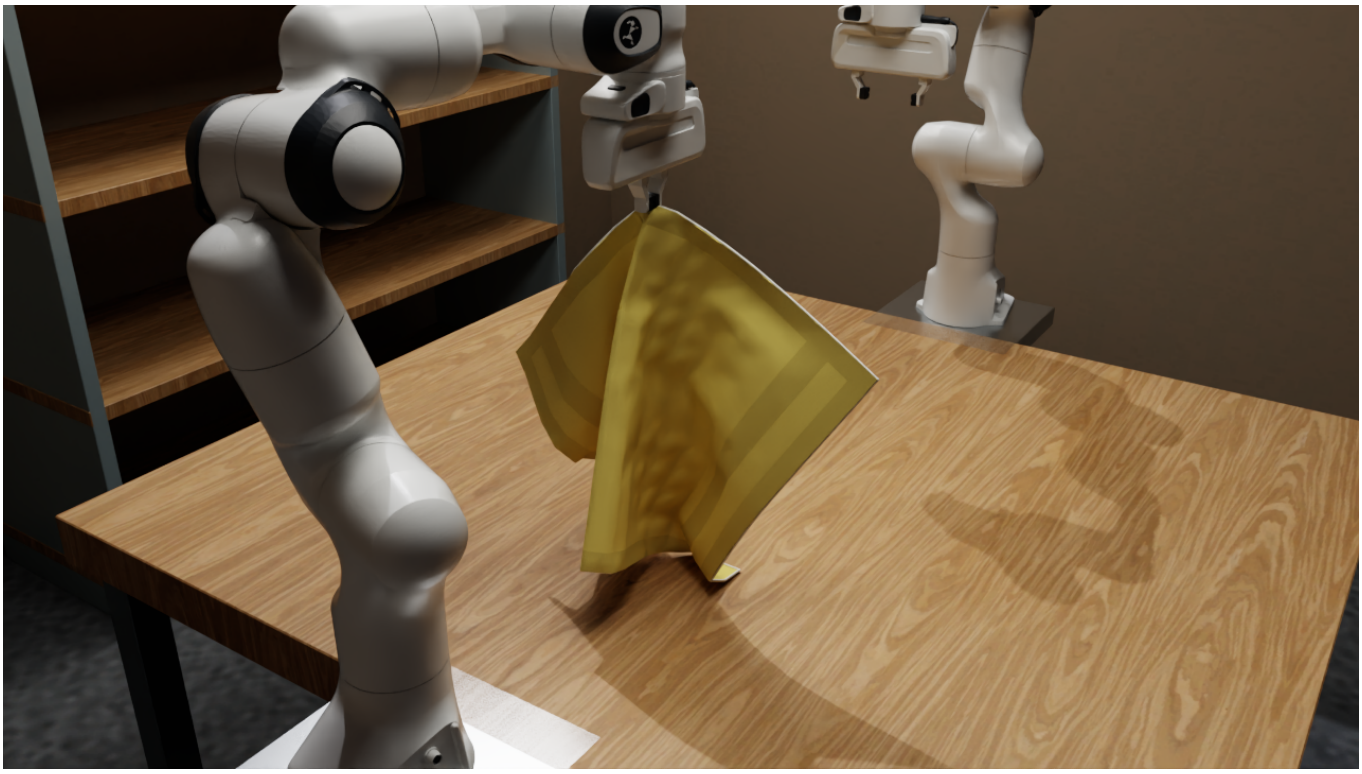


Figure 16: Paired source frame before normal interpolation: coarse-face shading patches. Same physical tick and geometry as the following view.

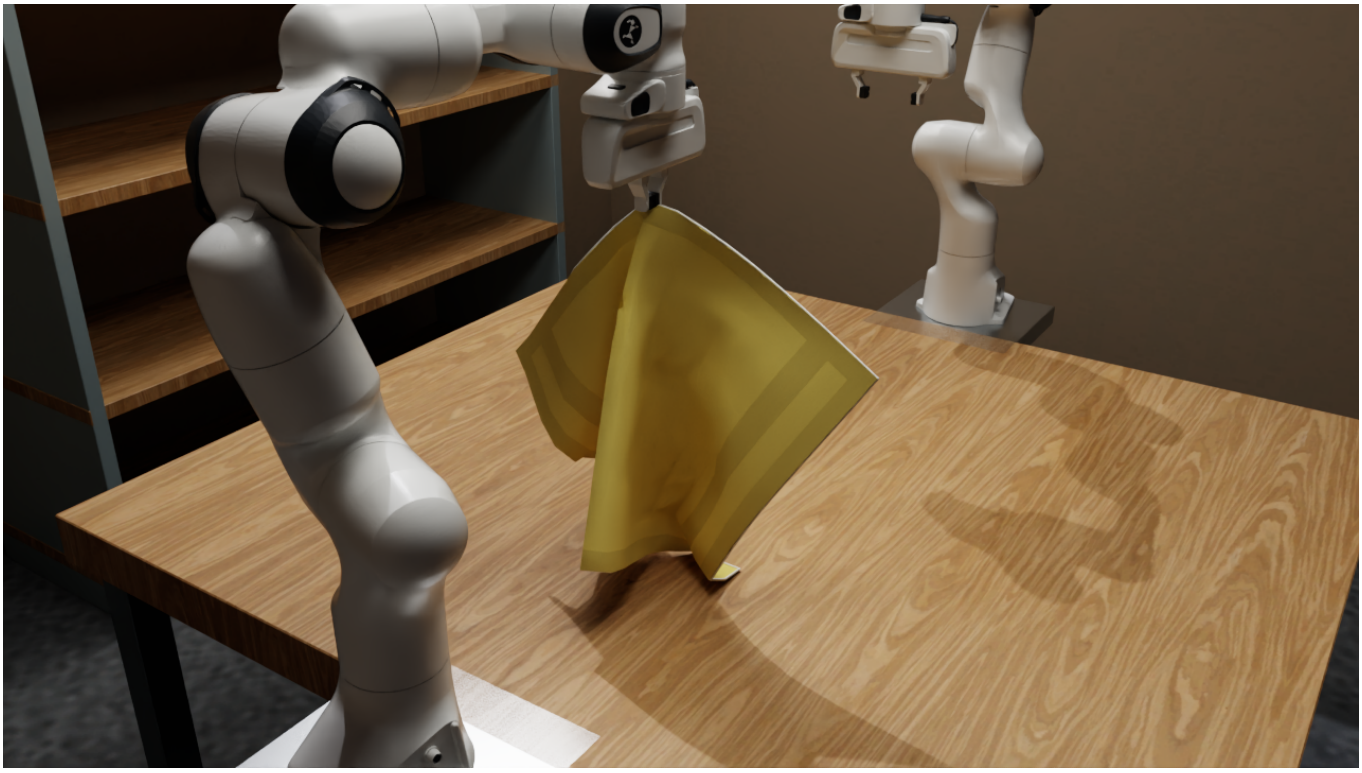


Figure 17: Interpolated panel normals reduce patchy highlights without changing physical or shell points. This does not add mechanical resolution.



Figure 18: Actual terry-loop close-up. Render-only loops follow material triangles; woven bands and hems are separate. There are no mechanical yarn attachments.

7.4 Camera roles and synchronized presentation

The policy receives the mounted external ideal-pinhole RGB sensor and named joint encoders. It is not a calibrated noisy depth camera. Diagnostic room, overhead, fabric and hand close-up cameras are privileged. Camera mounting, projection, timestamps and readback must be correct; seeing the entire task from every initial pose is not an acceptance condition.

The release videos provide an overview together with a useful closer view of the same recorded tick. Phase labels explain commanded operations, not assumed outcomes. Playback speed, simulation time, tick cadence and assistance are visible. Close-ups do not replace the context view or conceal a dropped cloth. Inspection includes first frame, grasp/support events, opening/release and final settling in the actual decoded MP4, not only stills created before encoding.

8 Tasks, observations and remote operation

8.1 Task definitions and evaluation

Garment	Existing construction-specific goal
T-shirt	Tuck both short sleeves into the body footprint.
Button shirt	Fold both long sleeves and cuffs onto the body.
Jeans	Fold one leg shell onto the other.
Terry towel	Fold long edge to long edge.
Asymmetric top	Extend both waist ties without intersections.

Table 18: These goals are not simplified to match the component demonstrations.

The semantic garment index maps named construction regions to physical particles. Evaluation uses measured geometry and task-specific scores plus integrity/release gates, including deformation, nonlocal intersections and absence of robot

contact when release is required. A component demonstration is not reported as a solved task merely because a grasp or release succeeded. The privileged evaluator is separate from policy camera observations.

8.2 Evaluator

For each goal the evaluator returns a geometric score s_k and named Boolean gates. The privileged step reward is the weighted mean $\Phi = \sum_k w_k s_k / \sum_k w_k$; success requires every gate. The button shirt is the exception: its score is exactly 1 when every gate passes and otherwise $0.9\,\Phi$ (0 with a confirmed cloth crossing), so every valid fold gets the same full credit and no unfinished state can tie it. This backend does *not* run the `GarmentEpisodeTask` potential-difference reward, grasp-history state machine or 60 s timeout. Restricted policy actions withhold reward and success. A high geometric score is not a verified release.

Let $S_-(v,t) = \mathrm{clip}(1 - v/(2t), 0, 1)$ and $S_+(v,t) = \mathrm{clip}(v/t, 0, 1)$. Alignment scores average S_- of the RMS and maximum distances; containment and extension use S_+. Stacking uses the mean of projected-nearest RMS and normal-gap scores. The table below states the actual gate thresholds from `garment_task.py`. They are task definitions, not textile material measurements.

Garment	All required geometric gates
T-shirt	Each sleeve: at least 90% of independent particles inside the live body convex hull (15 mm margin); nearest projected RMS $\leq$ 45 mm and 90th-percentile normal gap $\leq$ 25 mm.
Button shirt	For each side, at least 85% of the sleeve and 80% of the cuff (by rest area; sleeve cloth within 60 mm of the armhole seam excluded) lies on the body: over the projected body cloth (fronts, back, yoke, pocket, collar and stand; triangles, not a hull) within 20 mm and over the tabletop; resting with grounded support (at most 40 mm above the table, body cloth, or sleeve cloth that itself rests; the upper cloth layer of a flattened sleeve held up by its own lower layer does not rest); no front or back cloth both above and below it; at most seven cloth layers of its own sleeve and cuff in its column. The body stays laid out (hem-to-neckline distance at least 75% of the start's). See the shirt paragraph below.
Jeans	Symmetric nearest-particle projected RMS between leg shells $\leq$ 45 mm; 90th-percentile gap $\leq$ 30 mm. Front and back hem path RMS $\leq$ 25 mm and maximum $\leq$ 55 mm, using 17 arc-fraction samples.
Towel	Opposite long hems: tabletop-projected RMS $\leq$ 25 mm and maximum $\leq$ 60 mm, using 17 samples. Convex projected area $\leq$ 62% of the compiled assembly reference.
Asymmetric top	Each waist tie's anchor-to-tip distance is at least 85% of its compiled reference distance; endpoints average five path samples.

Table 19: The evaluator uses semantic particle groups and material paths, not render pixels or action names.

Every task also requires maximum particle speed ≤ 0.020 m/s, zero confirmed nonlocal intersections, and no robot-owned cloth contact witnesses at evaluation (the shirt's rest, strain and table gates are defined in its paragraph below). Allowed maximum absolute principal-stretch deviation is 0.30 for the tee (0.30 stretch only for the shirt), 0.25 for jeans, 0.20 for towel and 0.35 for the asymmetric top. Current-state values are checked, not historical peak strain. Public robot actions cannot write cloth positions or apply direct particle forces; the zero counters passed by this backend express that API restriction, not detection of arbitrary privileged Python code edits. The convex-hull and nearest-particle metrics (all tasks except the shirt) are explicit approximations to a visually recognizable fold.

Button shirt sleeve fold. The task text is: "Fold both long sleeves and cuffs onto the shirt body. Use the robot arms; leave the garment released and resting on the table." *Status: no fold of this sleeve has been shown to stay. In assisted (phantom-force) trials every folded sleeve, creased at the armhole or across the middle, pressed or not, sprang back off the body after release (an environment defect), and no Panda pinch has held the sleeve unassisted. Success has therefore been shown only on constructed states, and the task is not ready for policy evaluation.*

Start. The authored P13 shirt lay recoils (both cuffs move about 0.23 m within 0.07 s) and creeps for tens of seconds, so this task starts from the same shirt after 20 s untouched on the table (a stored at-rest lay: positions only, velocities zero, no lift). Its first image is the layout the robot acts on. The P13 lay remains available as `LaundryEnv(config,start="catalogue")`.

Success. Measured per unit of sleeve material, independent of face (front or behind the body), crossing order, fold line and method. Accepted outcomes checked on constructed states include the shirt-maker's guide fold (sleeve across,

folded up from mid-sleeve, cuff turned down), a sleeve folded in thirds, a doubled sleeve, a turned-back cuff, crossed sleeves and sleeves behind the body. Rejected by design: a sleeve laid inside the open front; a cuff tucked under the opposite front half (the natural straight fold carries each cuff across the opposite placket edge; the 20% cuff allowance tolerates its tip, about the last 2-3 cm, not a tucked cuff); the body folded over the sleeves (sides or hem folded over them, as the next steps of a full shirt fold do) or bunched below 75% of its laid-out length; sleeve material with more than seven cloth layers (levels at least 2 mm apart) of its own side's sleeve and cuff in its column, whether bunched or neatly folded. The guide fold and a sleeve folded in thirds have six, so one extra layer from a bulging crease is tolerated; a sleeve concertina'd into four or more full pleats has eight or more. Folding the body lengthwise *beneath* the sleeves (the retail side-in fold, or the shirt halved first) keeps the sleeves on top and passes. Further gates: *at rest*, no particle farther than 10 mm from its current position during the last 0.5 s (single-instant particle speeds of resting cloth show solver velocity noise of 0.1–2 m/s); *on the table*, at least 95% of the garment's rest area over the tabletop, not more than 10 mm below it and not more than 0.30 m above it; stretch only bounded (maximum principal stretch ≤ 1.30), because in-plane compression of a 40 mm triangle is how sub-mesh wrinkles and folds appear; release and zero crossings as for every task.

Score. Exactly 1 when every gate passes; 0 with a confirmed crossing; otherwise $0.9\,\Phi$. Φ averages the two sides; each side is 0.75 sleeve and 0.25 cuff, normalised so the start reads 0. Per unit: 1 on the body as above (on a body shortened toward half its length this falls linearly to 0.4); 0.4 if it rests over the body but is enclosed by body cloth or has more than seven own cloth layers in its column (brought to the body, not in the asked state); otherwise $0.5\,(1 - d/d_0)$ clipped to $[0, 0.5]$, where d is its 3D distance to the nearest body particle and d_0 that distance at the start (floor 50 mm), so cloth lifted straight up earns nothing and cloth in the air over the body earns less the higher it is; 0 off the tabletop (outside its rectangle, more than 10 mm below its surface or more than 0.30 m above it). No grasp, order, fold line or intermediate shape is rewarded.

8.3 Python creation and controls

Create through the unified package rather than a parallel laundry API:

```
import gym_anything_robotics as gar
from gym_anything_robotics.actions import NamedJointTarget, Gripper
env = gar.create("laundry", config={"garment": "towel"})
observation = env.obs()
state = env.get_state()       # privileged diagnostic state
env.act(NamedJointTarget({}, ticks=1440))
env.act(Gripper(arm="left", half_width_m=0.04, duration_s=0.5))
env.close()
```

Observing does not advance physics. Hold commands advance actual ticks under the existing physical actuator targets. Named joint and Cartesian commands are validated before stepping. IK computes joint targets but is not a collision-free motion planner. Durations must correspond to exact integer numbers of physics ticks; requested values are not silently rounded into a different experiment. Reset is an explicit trajectory discontinuity.

8.4 Authenticated HTTP

```
gar run laundry --task towel --port 8896
gar keys laundry
```

Use the session's distinct policy and owner bearer keys, which `gar keys laundry` prints. The role's `/v1/help` documents permitted routes. The policy requests observations and robot actions; the teacher additionally retrieves physical geometry, evaluation and history. Returned image URLs require authentication. Plain HTTP does not encrypt credentials; use a trusted network or TLS reverse proxy.

```
GET /v1/observation
Authorization: Bearer POLICY_KEY

POST /v1/action
Authorization: Bearer POLICY_KEY
Idempotency-Key: left-open-001
Content-Type: application/json

{"type":"gripper","arm":"left",
```

```
"half_width_m":0.04,"duration_s":0.5}
```

Reuse an idempotency key only for retrying the identical action request. A different physical command requires a new key. Operations are serialized on one persistent environment. A failure response must not be interpreted as proof that no physics occurred; use the recorded action/tick state and the gateway's documented failure semantics before retrying.

If the separate render worker has exited, one observation may rebuild that worker and repeat the image request at the unchanged physical tick. It does not reset the robot or garment. Recovery records include the exit code and tick in privileged state. At most one retry per observation and three renderer restarts per backend lifetime are allowed; an invalid request or a slow live worker is not treated as a crashed process.

8.5 Teacher history and storage

The action trace stores particle and robot poses/velocities, requested and applied targets, actuator effort, rigid/cloth contact witnesses, solved rigid forces, queued/applied cloth reaction wrenches, contact begin/end events and integrity diagnostics. The timing description distinguishes step-start geometry from final-iteration force sampling. Each action has an ID and start/end ticks; completeness and eviction are explicit.

Normal HTTP actions bracket traces automatically. Direct Python operators must explicitly begin/end recording if they require archived history. The live store is bounded to 256 MiB of payload and 100,000 sample/event items; image retention is the latest 64 files. Clients must export evidence they intend to keep. No unbounded per-tick image directory is presented as a safe production recording policy. Digital component recordings retain their own trajectories separately.

8.6 Known limitations, stated at the correct scope

The current model does not simulate textile moisture, laundering, plastic crease memory, seam rupture, opening fasteners or yarn-level mechanical contact. The robot–cloth bridge is lagged and the cloth/rigid contact sample spacing can miss narrow contact locations. IK is not collision planning. The camera is an ideal RGB sensor. These are concrete model boundaries; the absence of physical-world calibration is not a substitute explanation for any observed failure.

The edge-drape ledger reports local strain extrema as well as average area strain. In the jeans trial, the minimum principal stretch reaches about 0.50 while the mean absolute area strain is about 0.57 percent. This is localized triangle compression in a coarse sewn mesh, not uniform shrinking of the garment. It exceeds the task's settled-fold strain limit; the gravity/contact component pass is therefore not represented as satisfying that separate gate.

9 Reproduction and artifact provenance

Install the release, then start a garment task with `gar run laundry --task towel`. Newton 1.5 physics and Isaac Sim 5.1 rendering use different interpreters; the renderer never owns garment physics. Set `LAUNDRY_RENDER_PYTHON` to the local Isaac interpreter. The manufacturer robot resolver verifies its installed assets.

Each recorded experiment keeps its exact effective settings, trajectory hashes, image/video inputs, counts and continuity. The report builder reads those inputs and generates tables/plots without running physics. Missing evidence is a build error, not a blank section disguised as completion.

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